

Growing Your Savings: Understanding Compound Interest

MoneySmart Compound Interest Calculator · ASIC MoneySmart

Years 7-10	~30 minutes	Mathematics · Number, Ratios & Financial Maths
------------	-------------	--

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://moneysmart.gov.au/budgeting/compound-interest-calculator>

Curriculum links: AC9M7N06 · AC9M7N09 · AC9M10A04 · finance

Learning intentions

- I can use the compound interest calculator to model how savings grow over time.
- I can calculate percentage growth using efficient strategies.
- I can compare how starting amount, interest rate and time affect long-term savings, and evaluate the assumptions behind the model.

Getting started

Open the MoneySmart Compound Interest Calculator. Set up a savings goal you might realistically have (e.g. saving for a bike, phone or trip) by entering a starting amount, a regular monthly deposit, an interest rate and a time period.

Tasks

1. Enter a starting balance of \$500, no extra monthly deposits, an interest rate of 5% p.a., and a term of 5 years. Record the final balance shown.

2. Keeping everything else the same, change the interest rate to 8% p.a. How much more money do you end up with after 5 years?

3. Using your two results above, calculate the difference in dollars and as a percentage increase.

Working:

4. Keep the interest rate at 5% p.a. and change only the number of years (e.g. 5, 10, 20). Record the final balance for each.

Years	Final balance

5. Describe what happens to the balance as the number of years increases. Is the growth pattern a straight line or a curve? Explain why, using the word 'compound'.

6. Now add a regular monthly deposit of \$50 to your original 5-year, 5% scenario. How much extra does this add to your final balance compared to Task 1?

7. A friend says 'it doesn't matter when you start saving, as long as you save the same total amount.' Use the calculator to compare starting 5 years earlier with a smaller deposit versus starting later with a bigger deposit, and explain whether your friend is correct.

Extension (optional)

This calculator assumes a fixed interest rate for the whole period. Explain one way this assumption might not match real life, and how that could change your final balance.
